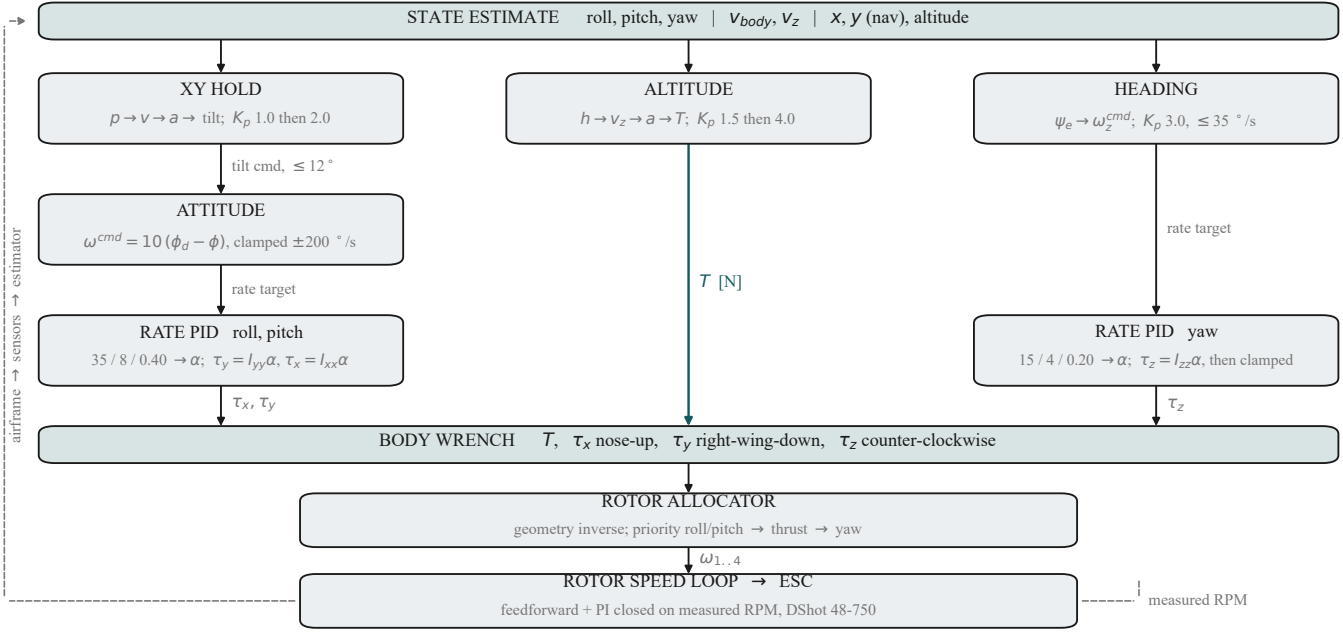

STATE ESTIMATE roll, pitch, yaw | v_{body}, v_z | x, y (nav), altitude
XY HOLD
$p \to v \to a \to$ tilt; K_p 1.0 then 2.0
ALTITUDE
$h \to v_z \to a \to T$; K_p 1.5 then 4.0
HEADING
$\psi_e \to \omega_z^{cmd}$; K_p 3.0, $\leq$ 35 °/s
tilt cmd, $\leq$ 12 °
ATTITUDE
$\omega^{cmd} = 10\,(\phi_d - \phi)$, clamped ±200 °/s
rate target
T [N]
rate target
RATE PID roll, pitch
35 / 8 / 0.40 $\to \alpha$; $\tau_y = I_{yy}\alpha$, $\tau_x = I_{xx}\alpha$
RATE PID yaw
15 / 4 / 0.20 $\to \alpha$; $\tau_z = I_{zz}\alpha$, then clamped
τ_x, τ_y
τ_z
BODY WRENCH T, τ_x nose-up, τ_y right-wing-down, τ_z counter-clockwise
ROTOR ALLOCATOR
geometry inverse; priority roll/pitch → thrust → yaw
$\omega_{1..4}$
ROTOR SPEED LOOP → ESC
feedforward + PI closed on measured RPM, DShot 48-750
measured RPM
airframe → sensors → estimator